

Git Commands in Windows (Step-by-Step Explanation)

1. Creating a Project Folder

`mkdir git-course` → Creates a new folder named git-course.

`cd git-course` → Moves into the git-course directory.

`dir` → Lists files and folders in Windows.

2. Windows vs Linux Commands

In Windows Command Prompt:

- `ls` and `cat` do NOT work (they are Linux commands).
- Use `dir` instead of `ls`.
- Use `type` instead of `cat`.

3. Creating and Viewing a File

`echo "git-course-demo" >> README.md` → Creates a file and writes content.

`type README.md` → Displays file content in Windows.

4. Initializing Git Repository

`git init` → Initializes a new Git repository (.git folder created).

`git add README.md` → Adds file to staging area.

`git commit -m "first commit"` → Saves changes to repository.

`git branch -M main` → Renames branch to main.

5. Generating SSH Key

`ssh-keygen -o` → Generates SSH public/private key pair.

Keys are stored inside `C:\Users\DELL\.ssh` folder.

`id_ed25519` → Private key (keep secret).

`id_ed25519.pub` → Public key (add to GitHub).

6. Connecting to GitHub using SSH

git remote add origin git@github.com:username/repository.git → Adds remote repository.
git push -u origin main → Pushes code to GitHub using SSH.
First time: You confirm GitHub authenticity by typing 'yes'.

Difference Between HTTPS and SSH in Git

Feature	HTTPS	SSH
Authentication	Username & Password / Token	SSH Key Pair (Public/Private)
Security	Secure (SSL/TLS)	Very Secure (Encrypted Key-Based)
Ease of Setup	Easy for beginners	Requires SSH key setup
Login Every Time?	May ask for credentials	No password after setup
Example URL	https://github.com/user/repo.git	git@github.com:user/repo.git

Conclusion

- Use HTTPS if you want simple setup.
- Use SSH if you want secure and password-free authentication.
- For regular developers, SSH is recommended after initial setup.